

G. Narayanamma Institute of Technology & Science

(Autonomous)

(for Women)

Shaikpet, Hyderabad- 500 104

IV-B.Tech I-Semester Supplementary Examinations, May-2026

DATA SCIENCE USING R
(Computer Science and Engineering)

Max. Marks: 60

Time: 03 Hours

Note:

1. Question paper comprises of **Part A** and **Part B**.
2. **Part A** (10 marks) is **compulsory** and consists of **ten sub-questions**, from **all units**, each carrying **1 mark**.
3. **Part B** (for 50 marks) consists of **five questions** with **“either” “or”** pattern. Each question carries 10 marks and may have a,b,c as sub questions. The student has to answer any one full question.

PART-A

(Answer 10 questions. Each question carries 1 mark)

Q.No.	Question	Marks	CO	BTL
Q.1	a) What are the roles and responsibilities of Data science project?	[01]	C01	[L1]
	b) What is the disadvantage of using R in enterprise-level large-scale solutions?	[01]	C01	[L1]
	c) How are the missing values in the input vector removed?	[01]	C02	[L1]
	d) Differentiate between invalid values and outliers.	[01]	C02	[L2]
	e) What are the various challenges that are to be addressed by Mapping problems to machine learning tasks?	[01]	C03	[L1]
	f) Compare Prediction and forecasting.	[01]	C03	[L2]
	g) What is a response and predictor variable?	[01]	C04	[L1]
	h) What is the difference between hierarchical and partitioning clustering strategies?	[01]	C04	[L2]
	i) What is decomposing in time series analysis?	[01]	C05	[L1]
	j) How does one analyse the autocorrelation and partial correlation for fitting an ARIMA model?	[01]	C06	[L1]

PART-B

(Answer 05 full questions. Each question carries 10 marks)

Q.No.	Question	Marks	CO	BTL
Q.2(a)	Discuss about the Data collection and management mechanisms of a Data science project.	[05]	CO1	[L1]
(b)	Describe some commonly used packages in R.	[05]	CO1	[L1]
OR				
Q.3(a)	Classify different data types in R with example. Explain how to check the data type of a variable and convert between different data types.	[05]	CO1	[L2]
(b)	With suitable example, explain Reading Data from Packages using R. What are the uses of <code>data()</code> , and <code>read.xlsx()</code> functions? Give their syntaxes.	[05]	CO1	[L3]

- Q.4(a) Explain RSQLite package. Create a dataset or table ['Shop'] and apply all the data exploring functions on this table. [04] CO2 [L4]
- (b) Write a R program to create a data frame using two given vectors and display the duplicated elements and unique rows of the data frame. Explain with syntax. [06] CO2 [L3]

OR

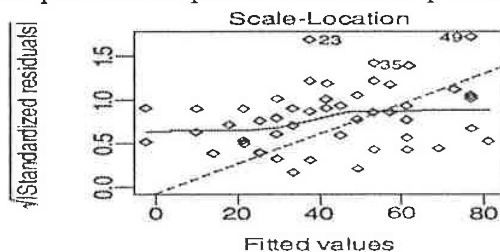
- Q.5(a) With suitable example and syntaxes, explain how to visually check relationships between two variables. [05] CO2 [L2]
- (b) Describe Creating and applying a simple treatment plan using *vtreat* package for automatically treating missing variables. [05] CO2 [L2]

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- Q.6(a) When to use Association rules? Describe Problem-to-method mapping with suitable example. [05] CO3 [L3]
- (b) What is k-fold cross-validation? With example explain Partitioning data for 3-fold cross-validation. [05] CO3 [L2]

OR

- Q.7(a) Suppose that you had multiple spam filters to choose from, each with different values of precision and recall. Explain how do you pick which spam filter to use? [04] CO3 [L2]
- (b) What is Sensitivity and Specificity? Suppose that you have successfully trained a spam filter with acceptable precision and recall, using your work email as training data. Now you want to use that same spam filter on a personal email account that you use primarily for your photography hobby. Will the filter work as well? Explain your answer. [06] CO3 [L4]

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- Q.8(a) Explain the use of *cor()* Function in R to Determine the Degree and Direction of Linear Association. [05] CO4 [L3]
- (b) Consider the "cars" data set. Assume "cars\$dist" as the response variable and "cars\$speed" as the predictor variable. Create a model using the *lm()* function. Explain the below plot with respect to residuals as per the model. [05] CO4 [L4]



OR

- Q.9(a) Explain binary logistic regression with a covariate variable. [05] CO4 [L2]
- (b) Create a matrix and implement k-means clustering on it. [05] CO4 [L3]

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- Q.10(a) Describe the basic R commands for the visualisation, and manipulation of time series data? [05] CO6 [L2]
- (b) Explain simple exponential smoothing with an example. [05] CO5 [L2]

OR

- Q.11(a) Explain the *acf()* function with syntax and an example. [05] CO6 [L2]
- (b) Create a simple time series data and decompose it. Interpret the output. [05] CO5 [L2]